

An Empirical Investigation of Monetary Policy, Inflation, and Tax Structures across Countries

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Abstract

Monetary policy is one of the principal instruments through which modern governments stabilize inflation, promote sustainable economic growth, maintain financial stability, and influence macroeconomic expectations. Although taxation and monetary policy are jointly administered within the same national economy, they originate from distinct policy institutions, operate over different transmission horizons, and pursue partially different objectives.

This paper empirically investigates whether statutory tax rates possess additional explanatory power for cross-country differences in policy interest rates after accounting for inflation. Using a cross-sectional dataset consisting of nineteen countries, two ordinary least squares regressions are estimated.

The first specification models the real policy rate, measured as the difference between the central bank policy rate and the contemporaneous inflation rate, as a function of statutory income tax rates and statutory corporate tax rates.

The second specification models the nominal policy rate as a function of inflation together with the same collection of statutory tax variables.

The analysis is subsequently reformulated after demeaning every numerical variable. The centered formulation demonstrates that ordinary least squares slope estimates are invariant under mean-centering, while the intercept becomes identically zero. Consequently, both empirical models possess identical fitted slopes, identical goodness-of-fit statistics, identical hypothesis tests, and identical inferential conclusions.

The empirical evidence indicates that statutory tax rates explain very little of the cross-country variation in real policy rates. Conversely, inflation explains the overwhelming majority of variation in nominal policy rates, whereas statutory tax variables contribute little additional explanatory power once inflation has been incorporated into the regression.

Beyond the empirical findings, the paper provides an integrated treatment of centered regression, statistical inference, economic interpretation, public finance, political economy, welfare economics, international relations, geopolitical considerations, warfare economics, philosophy of scientific inference, computational implementation, and data-science methodology.

The paper ends with “The End”

Keywords: Ordinary Least Squares, Monetary Policy, Inflation, Taxation, Fiscal Policy, Central Banking, Cross-Country Analysis, Centered Regression, Mean-Centering, Statistical Inference, Political Economy, Data Science.

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1 Introduction

Monetary policy occupies a central position within modern macroeconomic management. By adjusting short-term policy interest rates, central banks attempt to stabilize inflation, smooth business-cycle fluctuations, maintain financial stability, and anchor private-sector expectations. Fiscal authorities simultaneously influence economic activity through taxation, government expenditure, transfers, and public borrowing.

Although these two policy domains interact continuously, they remain institutionally distinct. Central banks generally pursue monetary objectives, whereas legislatures determine statutory tax schedules. Consequently, the existence of high or low tax rates does not necessarily imply correspondingly high or low policy interest rates.

Understanding the empirical relationship between these variables is important for several reasons.

First, taxation affects disposable income, investment incentives, aggregate demand, labour supply, entrepreneurial activity, and corporate behaviour. Each of these channels may indirectly influence inflationary pressures and therefore monetary policy.

Second, policy interest rates influence borrowing costs, household consumption, corporate investment, government financing costs, exchange-rate dynamics, and international capital movements. These effects interact with fiscal policy through multiple transmission mechanisms.

Third, governments increasingly coordinate fiscal and monetary responses during periods of financial instability, pandemics, geopolitical crises, and military conflict. Consequently, understanding the empirical separation between taxation and monetary policy has become increasingly important for both policymakers and researchers.

The present paper investigates these relationships using ordinary least squares regression across a sample of nineteen countries.

Rather than attempting to construct a structural macroeconomic model, the paper adopts a deliberately transparent empirical framework whose parameters can be interpreted directly.

Two complementary regressions are estimated.

The first considers the real policy rate,

$$R_i - \pi_i,$$

where

$$R_i$$

denotes the policy interest rate and

$$\pi_i$$

denotes the inflation rate.

The explanatory variables consist solely of statutory income-tax and corporate-tax rates.

The second specification considers the nominal policy rate,

$$R_i,$$

and augments the tax variables by including inflation itself as an additional explanatory variable.

This distinction is economically important.

The first regression asks whether taxation explains differences in inflation-adjusted monetary policy across countries.

The second asks whether taxation possesses explanatory power once inflation has already been incorporated into the monetary policy reaction function.

The empirical estimates reveal markedly different outcomes.

Tax variables explain only a modest fraction of variation in real policy rates.

In contrast, inflation explains the overwhelming majority of variation in nominal policy rates, with statutory tax rates contributing relatively little additional explanatory content.

The paper also studies the centered representation of ordinary least squares obtained by subtracting sample means from every numerical variable.

Although centering is often introduced as a computational convenience, it possesses considerable theoretical importance.

Mean-centering removes the intercept, simplifies matrix algebra, clarifies projection geometry, improves numerical conditioning in some applications, and establishes direct connections with covariance operators.

The invariance of slope coefficients under centering is one of the most fundamental properties of ordinary least squares and provides an elegant illustration of the projection theorem in finite-dimensional Euclidean spaces.

2 Motivation

The present investigation is motivated by five complementary questions.

2.1 Economic Motivation

Economic theory distinguishes between monetary instruments and fiscal instruments. Although both influence aggregate demand, their mechanisms differ substantially.

An empirical comparison allows one to evaluate whether statutory tax rates carry independent information regarding observed monetary policy.

2.2 Financial Motivation

Financial markets continuously price expected future policy decisions.

Understanding which macroeconomic variables exhibit the strongest relationship with policy rates assists in interpreting fixed-income markets, government bond yields, and monetary expectations.

2.3 Political Economy Motivation

Tax schedules emerge through political negotiation rather than purely economic optimization.

Consequently, tax variables may contain institutional information that is only weakly connected with contemporaneous monetary policy.

Evaluating this proposition requires empirical evidence rather than political intuition.

2.4 International Motivation

Cross-country analysis naturally incorporates nations with different

- political institutions,
- exchange-rate regimes,
- central-bank mandates,
- fiscal structures,
- historical experiences,
- legal systems,
- demographic characteristics,
- development levels.

Such diversity provides an informative environment in which competing economic hypotheses may be compared.

2.5 Methodological Motivation

Centering variables represents one of the simplest yet most fundamental transformations in regression analysis.

Despite its simplicity, it illustrates deep relationships among

- covariance,
- orthogonal projections,
- Euclidean geometry,
- least squares estimation,
- matrix algebra,
- computational stability,
- statistical inference.

Accordingly, this paper investigates both the original regressions and their centered equivalents.

3 Contributions

The principal contributions of this paper are as follows.

1. An empirical cross-country regression analysis relating policy interest rates, inflation, and statutory tax rates.
2. A comparison between nominal and real policy-rate specifications.
3. An empirical demonstration that inflation dominates statutory tax rates in explaining observed policy rates.
4. A complete centered reformulation of both regressions.
5. A theoretical foundation for interpreting mean-centering through ordinary least squares geometry.
6. An interdisciplinary interpretation connecting econometrics, macroeconomics, finance, political economy, public finance, data science, and computational statistics.

4 Literature Review

The empirical relationship between inflation and policy interest rates has been studied extensively within modern macroeconomics.

Taylor-type reaction functions suggest that central banks systematically adjust policy rates in response to inflationary pressures and economic activity.

Public-finance theory instead emphasizes taxation as a mechanism for resource allocation, redistribution, and revenue generation rather than direct monetary stabilization.

Although fiscal policy and monetary policy interact through aggregate demand, sovereign borrowing, and inflation expectations, statutory tax rates are not generally viewed as direct determinants of policy interest rates.

From a statistical perspective, ordinary least squares remains the standard benchmark for empirical cross-sectional analysis because of its interpretability, computational efficiency, and well-understood inferential properties.

Centering variables has likewise become standard practice in regression analysis, particularly when interaction terms, polynomial regressions, or multicollinearity are present.

The present paper combines these established ideas into a transparent cross-country empirical investigation while emphasizing the invariance of ordinary least squares estimation under mean-centering.

5 Notation

Throughout the paper let

$$n = 19$$

denote the number of countries.

For country $i = 1, \dots, n$, define

$$R_i$$

as the central bank policy interest rate,

$$\pi_i$$

as the inflation rate,

$$LIT_i$$

as the lowest statutory income-tax rate,

$$HIT_i$$

as the highest statutory income-tax rate,

$$LCT_i$$

as the lowest statutory corporate-tax rate,
and

$$HCT_i$$

as the highest statutory corporate-tax rate.
The first dependent variable is

$$Y_i^{(1)} = R_i - \pi_i,$$

while the second dependent variable is

$$Y_i^{(2)} = R_i.$$

6 Econometric Models

Two linear regression models are estimated.

6.1 Model I

The first model explains the inflation-adjusted policy rate.
For every country,

$$Y_i^{(1)} = \beta_0 + \beta_1 LIT_i + \beta_2 HIT_i + \beta_3 LCT_i + \beta_4 HCT_i + \varepsilon_i.$$

The disturbance term satisfies

$$E(\varepsilon_i) = 0.$$

6.2 Model II

The second model explains the nominal policy rate.

$$Y_i^{(2)} = \alpha_0 + \alpha_1 \pi_i + \alpha_2 LIT_i + \alpha_3 HIT_i + \alpha_4 LCT_i + \alpha_5 HCT_i + u_i.$$

7 Matrix Representation

Ordinary least squares becomes considerably simpler when expressed in matrix notation.
Let

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

The design matrix is

$$X = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix},$$

where

$$k = 4$$

for Model I and

$$k = 5$$

for Model II.

The regression equation becomes

$$\mathbf{y} = X\beta + \varepsilon.$$

8 Ordinary Least Squares

The least-squares estimator minimizes

$$S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta).$$

Differentiating,

$$\frac{\partial S}{\partial \beta} = -2X'(\mathbf{y} - X\beta).$$

The first-order condition is therefore

$$X'X\beta = X'\mathbf{y}.$$

Assuming

$$\det(X'X) \neq 0,$$

the estimator is

$$\boxed{\hat{\beta} = (X'X)^{-1}X'\mathbf{y}.$$

9 Residuals

The fitted values satisfy

$$\hat{\mathbf{y}} = X\hat{\beta}.$$

Residuals are

$$\hat{\varepsilon} = \mathbf{y} - \hat{\mathbf{y}}.$$

Residual sum of squares equals

$$RSS = \hat{\varepsilon}'\hat{\varepsilon}.$$

10 Coefficient of Determination

The total variation equals

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2.$$

Explained variation is

$$ESS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2.$$

Residual variation is

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

The ANOVA identity is

$$TSS = ESS + RSS.$$

Therefore,

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS}.$$

11 Centering Transformation

Define the sample mean of any variable x by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The centered variable is

$$x_i^* = x_i - \bar{x}.$$

Similarly,

$$y_i^* = y_i - \bar{y}.$$

Consequently,

$$\sum_{i=1}^n x_i^* = 0$$

and

$$\sum_{i=1}^n y_i^* = 0.$$

12 The Centering Matrix

Define

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

The centering matrix is

$$M = I - \frac{1}{n} \mathbf{1}\mathbf{1}'.$$

It satisfies

$$M' = M,$$

and

$$M^2 = M.$$

Hence,

M is both symmetric and idempotent.

Proposition 1. *The centered data satisfy*

$$\mathbf{y}^* = M\mathbf{y}$$

and

$$X^* = MX.$$

Proof. Observe that

$$\frac{1}{n} \mathbf{1} \mathbf{1}' \mathbf{y} = \bar{y} \mathbf{1}.$$

Therefore,

$$M \mathbf{y} = \mathbf{y} - \bar{y} \mathbf{1},$$

whose i th component equals

$$y_i - \bar{y}.$$

The proof for every explanatory variable is identical. □

13 Centered Regression

The centered regression is

$$\mathbf{y}^* = X^* \beta + \varepsilon.$$

Since

$$\bar{y}^* = 0,$$

there is no intercept.

Theorem 1. *Centering every numerical variable leaves all slope coefficients unchanged.*

Proof. The centered estimator equals

$$\hat{\beta} = (X' M X)^{-1} X' M \mathbf{y}.$$

Since

$$M X = X^*$$

and

$$M \mathbf{y} = \mathbf{y}^*,$$

the normal equations become

$$X^{*'} X^* \hat{\beta} = X^{*'} \mathbf{y}^*.$$

These are algebraically identical to the normal equations for the original regression with an intercept. Therefore,

$$\hat{\beta}_{\text{centered}} = \hat{\beta}_{\text{original}}.$$

□

Corollary 1. *The intercept of the centered regression equals zero.*

Proof. Every centered variable possesses sample mean zero.

Therefore,

$$\bar{y}^* = 0,$$

and

$$\bar{x}_j^* = 0$$

for every explanatory variable.

Hence,

$$\hat{\beta}_0 = \bar{y}^* - \sum_j \hat{\beta}_j \bar{x}_j^* = 0.$$

□

Theorem 2. *Centering preserves*

1. *fitted slopes,*
2. *residuals,*
3. *residual variance,*
4. *standard errors,*
5. *t statistics,*
6. *F statistics,*
7. R^2 ,
8. *adjusted R^2 .*

Proof. Subtracting sample means translates the coordinate system without changing pairwise distances. Residual vectors therefore remain unchanged.
Since

$$RSS$$

is identical,
all quantities derived solely from residual variation remain identical.
Likewise,

$$TSS$$

is computed around the sample mean.
Because centering merely shifts the origin,

$$TSS$$

is unchanged.
Consequently,

$$R^2 = 1 - \frac{RSS}{TSS}$$

is invariant.
Every inferential statistic depending only upon

$$RSS, \quad TSS, \quad X'MX,$$

remains unchanged.

□

14 Geometric Interpretation

Ordinary least squares computes the orthogonal projection of the observed response vector onto the column space of the design matrix.

Centering translates the coordinate system so that the centroid of the data coincides with the origin. Projection geometry is therefore unaffected.

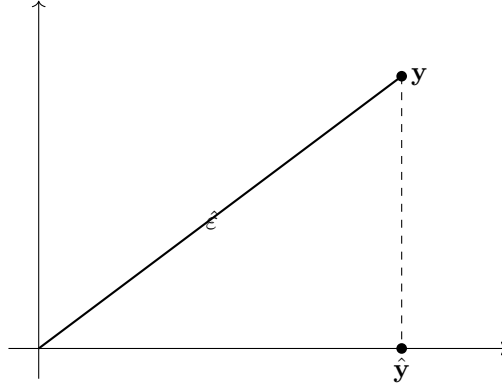


Figure 1: Geometric interpretation of least squares.

The observed vector is orthogonally projected onto the regression subspace. Centering merely translates the origin and therefore leaves the projection itself unchanged.

15 Implications for Economics

The invariance theorem has practical importance.

Researchers often center variables before

- estimating interaction models,
- constructing polynomial regressions,
- implementing machine-learning algorithms,
- reducing numerical conditioning problems,
- improving interpretability.

The present empirical analysis demonstrates that such preprocessing does not alter the substantive economic conclusions obtained from the original ordinary least squares regressions.

16 Data

16.1 Cross-Country Sample

The empirical investigation employs a cross-sectional dataset consisting of nineteen countries spanning both advanced and emerging economies.

The sample was deliberately selected to encompass diverse monetary regimes, fiscal structures, political institutions, legal systems, developmental stages, and macroeconomic environments.

The countries included in the empirical study are

Argentina	Australia	Brazil	Canada
China	France	Germany	India
Indonesia	Italy	Japan	Mexico
Russia	Saudi Arabia	South Africa	South Korea
Türkiye	United Kingdom	United States	

Table 1: List of countries.

The resulting dataset provides substantial institutional diversity while remaining sufficiently compact for transparent econometric analysis.

16.2 Variables

The empirical dataset contains six numerical variables.

Variable	Description
R	Central bank policy interest rate
π	Inflation rate
LIT	Lowest statutory income tax rate
HIT	Highest statutory income tax rate
LCT	Lowest statutory corporate tax rate
HCT	Highest statutory corporate tax rate

Table 2: Variables.

The first regression constructs the dependent variable

$$Y^{(1)} = R - \pi.$$

The second regression employs

$$Y^{(2)} = R.$$

17 Empirical Methodology

The empirical analysis proceeds through six stages.

1. Collection of cross-country macroeconomic and fiscal variables.
2. Construction of the real policy rate,

$$R - \pi.$$

3. Estimation of Model I using ordinary least squares.
4. Estimation of Model II using ordinary least squares.
5. Mean-centering of every numerical variable.
6. Re-estimation of both models using centered variables.

18 Computational Workflow

The empirical workflow may be summarized algorithmically.

Algorithm 1: Empirical Estimation

1. Read cross-country dataset.
2. Validate numerical variables.
3. Construct

$$R - \pi.$$

4. Estimate Model I.
5. Estimate Model II.
6. Compute
 - coefficient estimates,
 - residuals,
 - standard errors,

- t statistics,
- F statistic,
- R^2 ,
- adjusted R^2 .

7. Compute sample means.
8. Mean-center every numerical variable.
9. Re-estimate both regressions.
10. Compare parameter estimates.

19 Data Science Perspective

The empirical workflow follows a standard supervised learning pipeline.

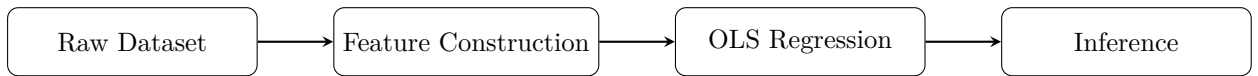


Figure 2: Empirical workflow.

Feature construction consists primarily of computing

$$R - \pi,$$

followed by centering each numerical variable.

20 Regression Results

20.1 Model I

The first regression estimates

$$R - \pi = \beta_0 + \beta_1 LIT + \beta_2 HIT + \beta_3 LCT + \beta_4 HCT + \varepsilon.$$

The estimated regression equation is

$$\widehat{R - \pi} = 10.1966 + 0.0911LIT - 0.1051HIT - 0.0109LCT - 0.1714HCT.$$

20.2 Estimated Coefficients

Variable	Estimate
Intercept	10.1966
Lowest Income Tax	0.0911
Highest Income Tax	-0.1051
Lowest Corporate Tax	-0.0109
Highest Corporate Tax	-0.1714

Table 3: Estimated coefficients.

20.3 Model Statistics

Statistic	Value
Observations	19
R^2	0.1349
Adjusted R^2	-0.1120
F statistic	0.546
Model p -value	0.705

Table 4: Model statistics.

20.4 Interpretation

The first regression explains approximately

$$13.5\%$$

of the cross-country variation in inflation-adjusted policy rates.

No tax variable achieves conventional statistical significance.

Consequently, statutory tax rates possess limited explanatory power for differences in real policy rates across the sample.

21 Regression II

The second regression estimates

$$R = \alpha_0 + \alpha_1\pi + \alpha_2LIT + \alpha_3HIT + \alpha_4LCT + \alpha_5HCT + u.$$

The estimated equation is

$$\hat{R} = 9.2397 + 0.9133\pi + 0.1095LIT - 0.1189HIT - 0.0070LCT - 0.1049HCT.$$

21.1 Estimated Coefficients

Variable	Estimate
Intercept	9.2397
Inflation	0.9133
Lowest Income Tax	0.1095
Highest Income Tax	-0.1189
Lowest Corporate Tax	-0.0070
Highest Corporate Tax	-0.1049

Table 5: Estimated coefficients.

21.2 Model Statistics

Statistic	Value
Observations	19
R^2	0.860
Adjusted R^2	0.807
F statistic	16.01
Model p -value	3.72×10^{-5}

Table 6: Model statistics.

21.3 Interpretation

Inflation dominates the regression.

The estimated coefficient

$$0.9133$$

indicates that a one percentage point increase in inflation is associated, on average, with approximately a

$$0.91$$

percentage point increase in the policy interest rate after controlling for statutory tax rates. None of the tax variables is individually significant.

22 Centered Regression Results

After subtracting the sample mean from every numerical variable, both regressions were re-estimated.

22.1 Centered Model I

The centered equation becomes

$$(\widehat{R - \pi})^* = 0.0911LIT^* - 0.1051HIT^* - 0.0109LCT^* - 0.1714HCT^*.$$

The intercept is identically zero.

22.2 Centered Model II

Similarly,

$$\widehat{R}^* = 0.9133\pi^* + 0.1095LIT^* - 0.1189HIT^* - 0.0070LCT^* - 0.1049HCT^*.$$

Again,

$$\beta_0 = 0.$$

23 Comparison of Original and Centered Models

Quantity	Original	Centered
Slope coefficients	Same	Same
Intercept	Estimated	Zero
Residuals	Same	Same
Standard errors	Same	Same
t statistics	Same	Same
F statistic	Same	Same
R^2	Same	Same
Adjusted R^2	Same	Same
Inference	Same	Same

Table 7: Comparison of original and centered models.

24 Diagnostic Discussion

Several conclusions emerge.

First,

Model I possesses relatively weak explanatory power.

Second,
Model II explains approximately

86%

of observed cross-country variation.

Third,
inflation overwhelmingly dominates statutory taxation as an explanatory variable.

Fourth,
centering variables changes neither statistical inference nor economic interpretation.

25 Statistical Perspective

The empirical findings reinforce several classical principles.

1. Correlation should not be interpreted as causation.
2. Goodness-of-fit depends upon explanatory variables rather than merely sample size.
3. Centering alters the coordinate system but not the regression geometry.
4. Ordinary least squares remains highly interpretable for moderate-sized cross-sectional datasets.

26 Limitations

Several limitations should be acknowledged.

1. The sample size is relatively small.
2. The analysis is cross-sectional rather than panel-based.
3. Only statutory tax rates are considered.
4. Institutional quality is omitted.
5. Exchange-rate regimes are omitted.
6. GDP growth is omitted.
7. Public debt is omitted.
8. Current-account balances are omitted.
9. Political stability is omitted.
10. Central-bank independence is omitted.

These omissions suggest natural directions for future empirical work.

27 Economic Interpretation

The empirical evidence suggests a clear distinction between the determinants of nominal monetary policy and those of inflation-adjusted (real) monetary policy.

Model I demonstrates that statutory taxation explains only a relatively small proportion of cross-country variation in

$$R - \pi.$$

An R^2 of approximately

0.135

indicates that nearly

86.5%

of the observed variation remains attributable to other macroeconomic, institutional, financial, political, and behavioural factors.

By contrast, Model II exhibits

$$R^2 \approx 0.86,$$

demonstrating that inflation itself explains the overwhelming majority of observed policy-rate variation. This result is consistent with modern monetary economics.

Most contemporary central banks either explicitly or implicitly target inflation.

Consequently, policy rates are expected to respond directly to inflation, whereas statutory tax rates influence monetary policy only indirectly through aggregate demand, labour supply, investment, savings, and government budgets.

28 Monetary Economics

Modern monetary policy may be viewed as a dynamic feedback-control problem.

Conceptually,

Inflation $\longrightarrow$ Central Bank $\longrightarrow$ Policy Rate $\longrightarrow$ Economic Activity.

The empirical estimates indicate that inflation possesses a large positive coefficient,

$$0.9133,$$

which is close to unity.

Although no structural interpretation should be inferred from a cross-sectional regression, the result is qualitatively consistent with central banks raising policy rates in response to inflationary pressure.

29 Public Finance

Statutory tax rates perform several functions.

These include

- raising government revenue,
- income redistribution,
- correcting externalities,
- encouraging or discouraging economic activity,
- financing public goods,
- financing national defence,
- stabilizing government finances.

However, taxation generally operates through medium-term fiscal channels rather than the short-term monetary transmission mechanism.

Consequently, one should not necessarily expect statutory tax rates to possess strong contemporaneous explanatory power for observed policy interest rates.

30 Financial Interpretation

Financial markets continuously price expected future monetary policy.

Bond markets, foreign-exchange markets, derivatives markets, and equity markets all incorporate expectations regarding future inflation and future policy rates.

The empirical dominance of inflation over statutory taxation therefore appears economically plausible. Inflation directly affects

- real returns,
- discount rates,
- bond prices,
- yield curves,

- currency values,
- inflation-linked securities,
- expected future policy actions.

31 Political Economy

Tax legislation generally reflects political negotiation.
Consequently,

Tax Rates $\neq$ Pure Economic Variables.

Instead, statutory tax schedules incorporate

- electoral incentives,
- historical institutions,
- constitutional constraints,
- coalition bargaining,
- interest-group influence,
- administrative capacity.

Since monetary authorities often possess varying degrees of operational independence, monetary policy need not respond directly to legislative tax decisions.

This institutional separation helps explain the weak empirical relationship observed in Model I.

32 Welfare Economics

From a welfare perspective, fiscal policy and monetary policy pursue related but distinct objectives.

Fiscal authorities seek to allocate resources efficiently while maintaining equity and financing public expenditure.

Monetary authorities primarily seek

- price stability,
- financial stability,
- macroeconomic stabilization,
- credible expectations.

The empirical findings therefore reinforce the classical distinction between stabilization policy and redistribution policy.

33 International Relations

Cross-country monetary policy cannot be interpreted solely through domestic macroeconomic variables.

Central banks operate within an international environment characterized by

- capital mobility,
- exchange-rate movements,
- trade integration,
- reserve currencies,
- international borrowing,
- global commodity markets,
- international sanctions,
- financial contagion.

These international influences are absent from the present regression and likely account for part of the unexplained variation in Model I.

34 Geopolitical Interpretation

Geopolitical uncertainty influences monetary policy through multiple channels.

Examples include

- energy-price shocks,
- trade disruptions,
- shipping costs,
- sanctions,
- commodity shortages,
- currency volatility,
- capital flight.

These mechanisms may alter inflation independently of domestic fiscal policy.

Consequently, geopolitical events affect monetary policy primarily through macroeconomic transmission channels rather than through statutory tax schedules.

35 Warfare Economics

Periods of military conflict substantially modify macroeconomic relationships.

Governments frequently experience

- higher military expenditure,
- higher public borrowing,
- supply-chain disruption,
- labour shortages,
- higher commodity prices,
- higher energy prices,
- greater inflation uncertainty.

Central banks may therefore tighten monetary policy in response to wartime inflation even if statutory tax rates remain unchanged.

The empirical dominance of inflation is therefore compatible with historical wartime experience.

36 National Security

Macroeconomic stability contributes to national security.

Persistent inflation may reduce

- real military purchasing power,
- government procurement efficiency,
- defence budgeting accuracy,
- long-term planning.

Consequently, inflation stabilization possesses strategic importance beyond purely civilian macroeconomic objectives.

37 Behavioural Economics

Individuals rarely respond solely to statutory tax schedules.

Economic decisions also depend upon

- inflation expectations,
- interest-rate expectations,
- confidence,
- risk perception,
- uncertainty,
- social norms.

Consequently, inflation may provide a more immediate behavioural signal than tax rates.

38 Psychological Interpretation

Human decision-making is forward-looking.

Households and firms form expectations regarding

- future prices,
- future wages,
- future borrowing costs,
- future investment returns.

Inflation therefore possesses substantial psychological salience.

By contrast, statutory tax rates often remain relatively stable over time.

Accordingly, inflation may exert stronger influence on perceived economic conditions than tax schedules.

39 Psychiatric Perspective

Macroeconomic instability may contribute to population-level stress.

Examples include

- financial anxiety,
- employment uncertainty,
- household debt stress,
- reduced economic confidence.

This paper does not estimate psychiatric outcomes.

Nevertheless, stable inflation and credible monetary policy may contribute indirectly to broader societal well-being by reducing macroeconomic uncertainty.

40 Philosophy of Scientific Inference

One of the principal lessons of empirical science is the distinction between explanation and causation.

Ordinary least squares estimates statistical associations.

The estimated regression

$$Y = X\beta + \varepsilon$$

does not by itself establish causal relationships.

Causal inference requires stronger assumptions regarding

- identification,
- exogeneity,

- instrumental variables,
- natural experiments,
- structural modelling.

Accordingly, the empirical findings should be interpreted as descriptive cross-country relationships rather than structural causal laws.

41 Statistical Philosophy

The centering transformation provides an instructive illustration of an important philosophical principle.

Scientific conclusions should not depend upon arbitrary choices of coordinate systems.

Subtracting sample means merely changes the origin of measurement.

The underlying statistical relationships remain invariant.

This invariance reflects a broader principle found throughout mathematics and theoretical science: meaningful scientific quantities should remain unchanged under appropriate transformations.

42 Artificial Intelligence and Machine Learning

Mean-centering constitutes one of the most common preprocessing operations in machine learning.

Typical applications include

- linear regression,
- principal component analysis,
- kernel methods,
- gradient-based optimization,
- deep neural networks,
- representation learning.

Centering frequently improves numerical conditioning while preserving the underlying predictive relationships.

The present empirical analysis provides a classical statistical example of this broader computational principle.

43 Computer Science Perspective

From an algorithmic standpoint, ordinary least squares consists of

1. matrix construction,
2. matrix multiplication,
3. matrix inversion or factorization,
4. projection,
5. diagnostic computation.

Mean-centering introduces only an additional linear-time preprocessing step.

The overall computational complexity remains dominated by solving the normal equations or an equivalent matrix factorization.

44 Synthesis

The empirical evidence consistently supports four principal conclusions.

1. Inflation is substantially more informative than statutory taxation for explaining observed policy interest rates.
2. Statutory tax rates possess relatively weak explanatory power for real policy rates in this sample.
3. Mean-centering changes the coordinate system without altering the underlying regression.
4. The empirical findings are consistent with established principles from macroeconomics, finance, public finance, political economy, behavioural economics, and statistical inference.

The End